Lecture 2
CO_2 Transport and Storage

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Outline

- CO_2 transport
- CO_2 storage
- International CO_2 storage activities
- CO_2 storage R&D in China

CO_2 transport

- Overview
- Expanding CO_2 transportation networks管网
- Pipeline design – the ROAD project设计
- Permitting CO_2 transport infrastructure – standards and codes标准规范
- Financial and commercial models for CO_2 transportation networks投资金融
- Recommendation and outlook

The world's first commercial scale carbon

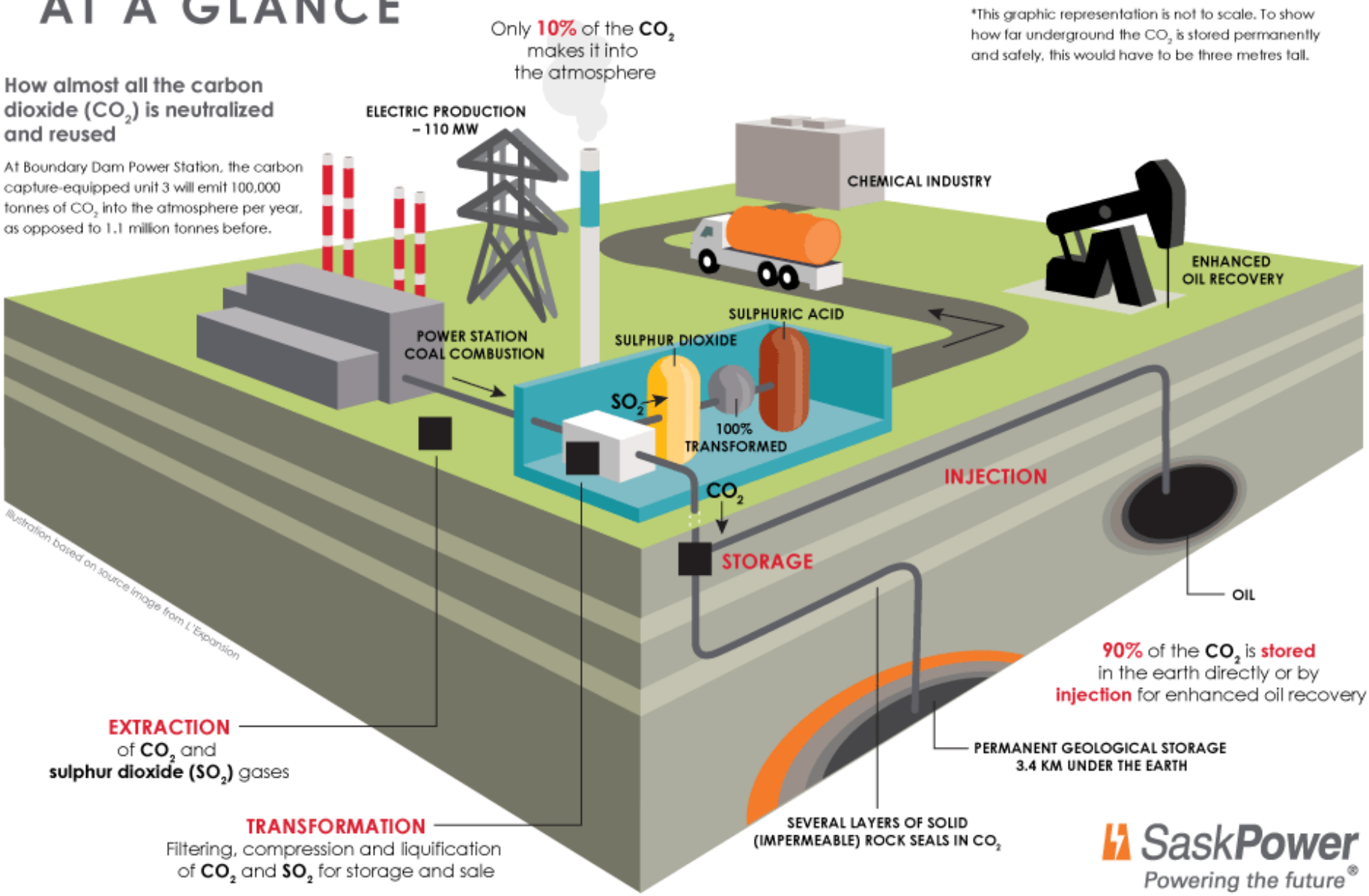
Carbon capture and storage

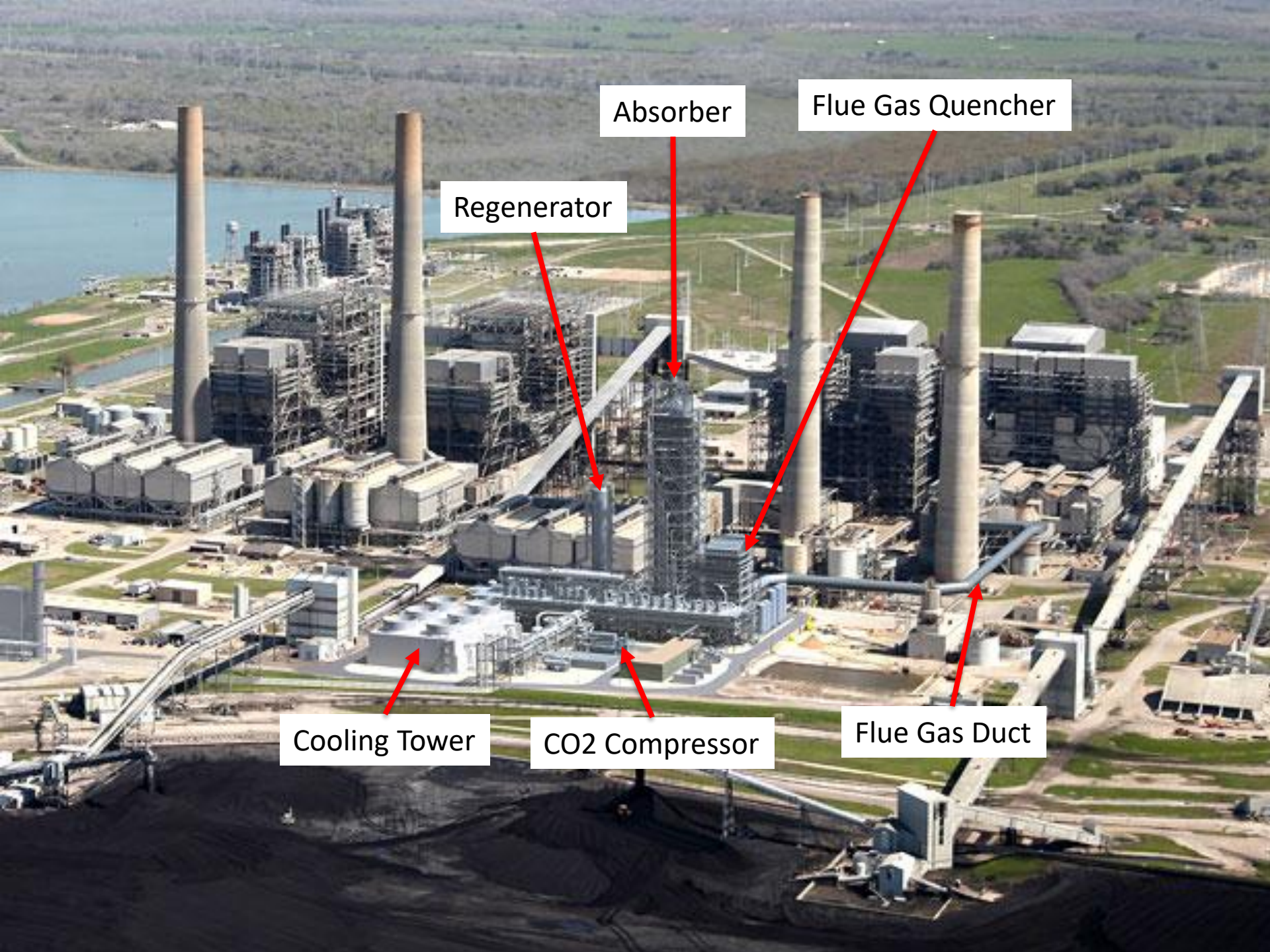
AT A GLANCE

How almost all the carbon dioxide (CO_2) is neutralized and reused

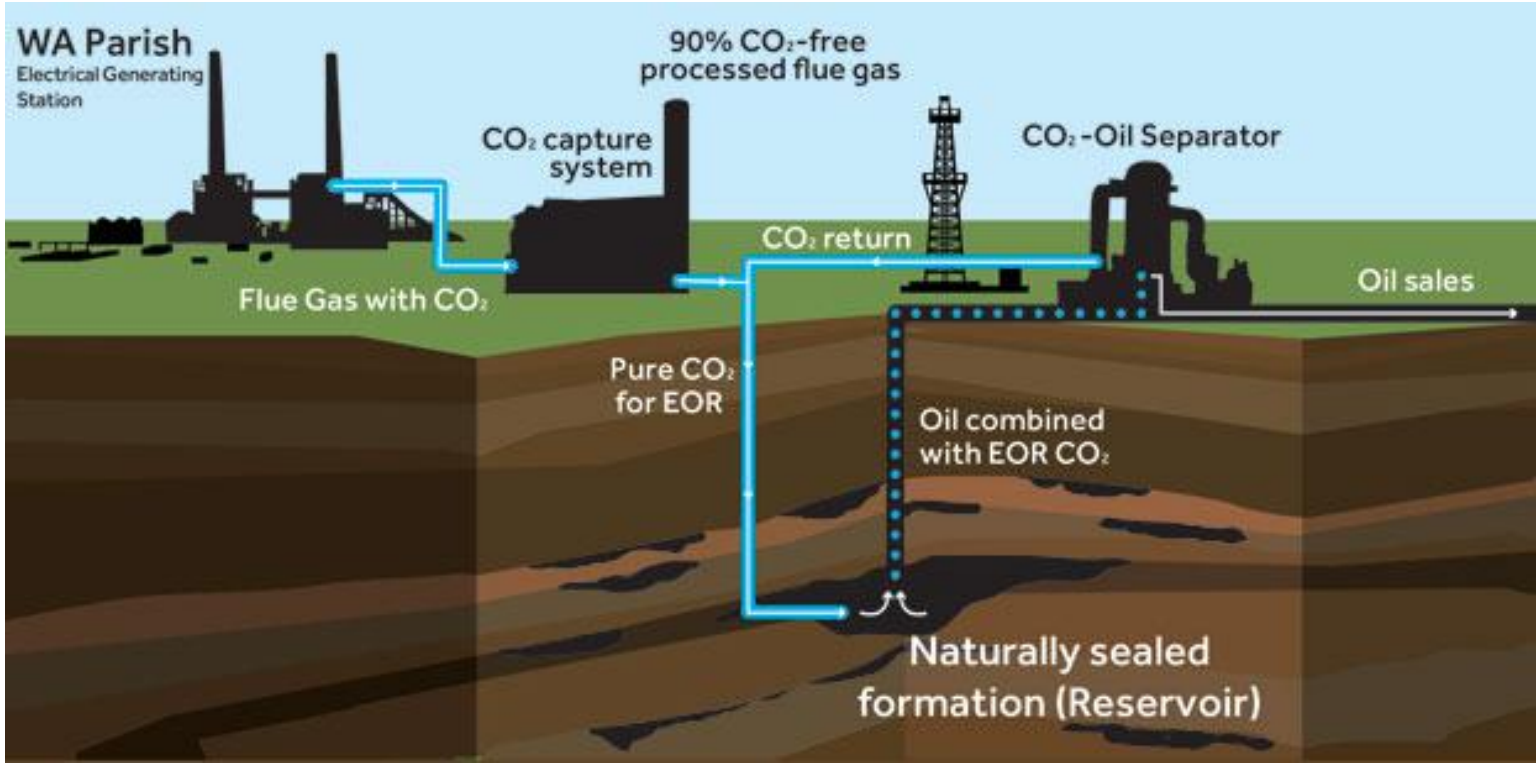
At Boundary Dam Power Station, the carbon capture-equipped unit 3 will emit 100,000 tonnes of CO_2 into the atmosphere per year, as opposed to 1.1 million tonnes before.

*This graphic representation is not to scale. To show how far underground the CO_2 is stored permanently and safely, this would have to be three metres tall.




Absorber
Flue Gas Quencher
Regenerator
Cooling Tower
CO2 Compressor
Flue Gas Duct

Petra Nova project



https://spectrum.ieee.org/energywise/green-tech/clean-coal/construction-starts-on-worlds-largest-postcombustion-carbon-capture-project

- The CO2 that is left over is compressed, and then sent via pipeline to the West Ranch oil field about 80 miles away. There, it will be used for enhanced oil recovery, which raises some questions about the actual environmental benefit involved here.
- 1.4 million metric tons of CO2 will be captured every year once it is up and running; it is estimated that the West Ranch oil field has about 60 million barrels of oil recoverable using enhanced extraction techniques, and production will be boosted from 500 barrels per day up to 15,000 per day, or more than 5 million per year.

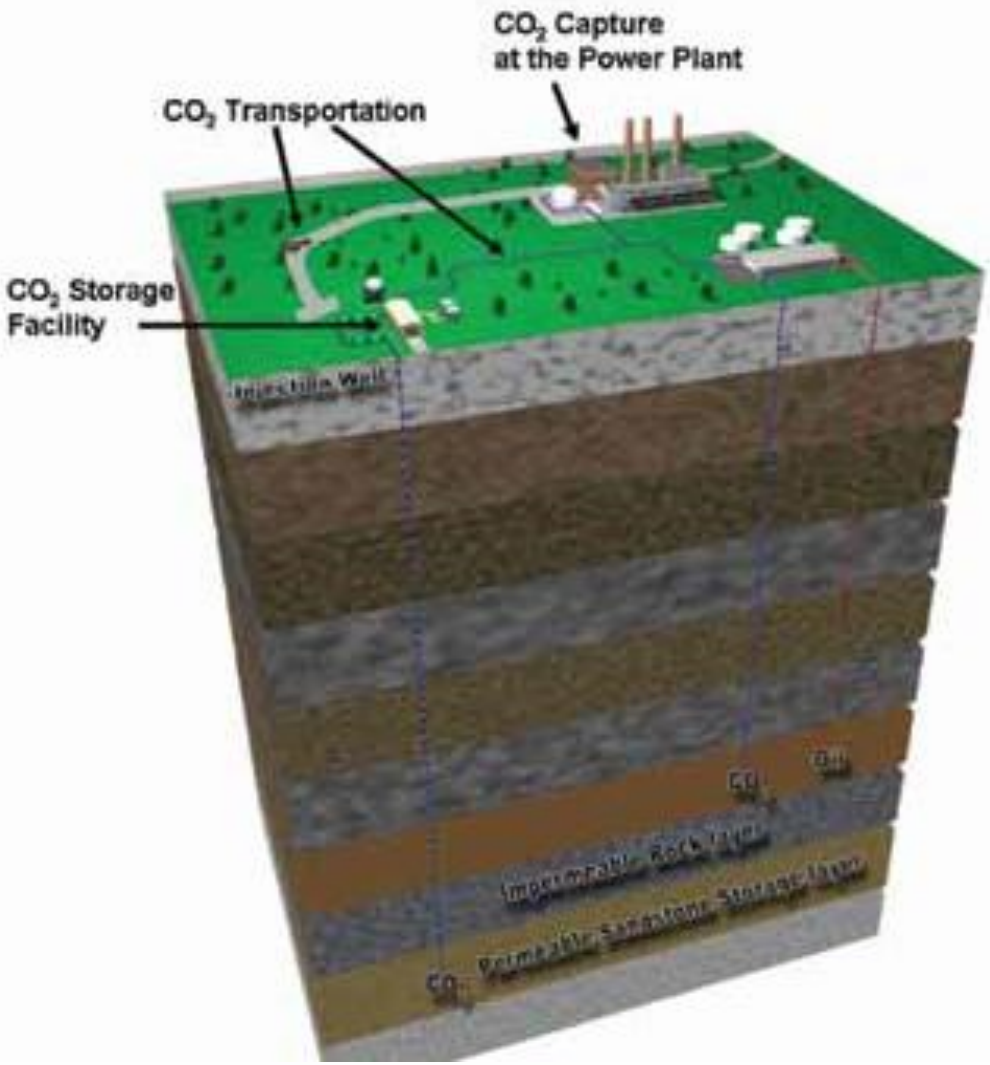


Figure 1-3. Carbon Capture and Storage Process

Source: 2010-DOE-NETL CO2 capture and storage RD&D Roadmap

CCS steps

CCS includes four primary steps:

- CO_2 capture,
- CO_2 compression,
- CO_2 transport,
- CO_2 storage.

The captured CO_2 could be transported via pipeline or tanker car to a permanent storage site. The CO_2 would then be stored underground in geologic formations such as depleted oil and gas fields, saline formations, and unmineable coal seams.

CO_2 Compression

- CO_2 captured from power plant will need to be compressed from near atmospheric pressure to a pressure between 1500 and 2200 psia (CO_2 is either in a liquid or supercritical phase) in order to be transported via pipeline and then injected into an underground storage site.
- The compression of CO_2 represents a potentially large auxiliary power load on the overall power plant system(about 7.5% of the gross power output of a subcritical pressure coal-fired power plant). The capital cost fro the compressor and associated equipment is also significant.

CO_2 transportation

- For most CCS applications, after CO_2 is captured and compressed, it needs to be transported to a suitable geologic storage site. Due to the large volumes of CO_2, transportation will likely be accomplished via pipeline rather than ground transportation.
- These pipelines will have potential impacts on numerous stakeholders, including land owners, nearby residents, pipeline companies, owners of storage site, power plants, environmental groups, and others who might be interested in a pipeline up to hundreds of miles long and carrying a pressurized / supercritical liquid.

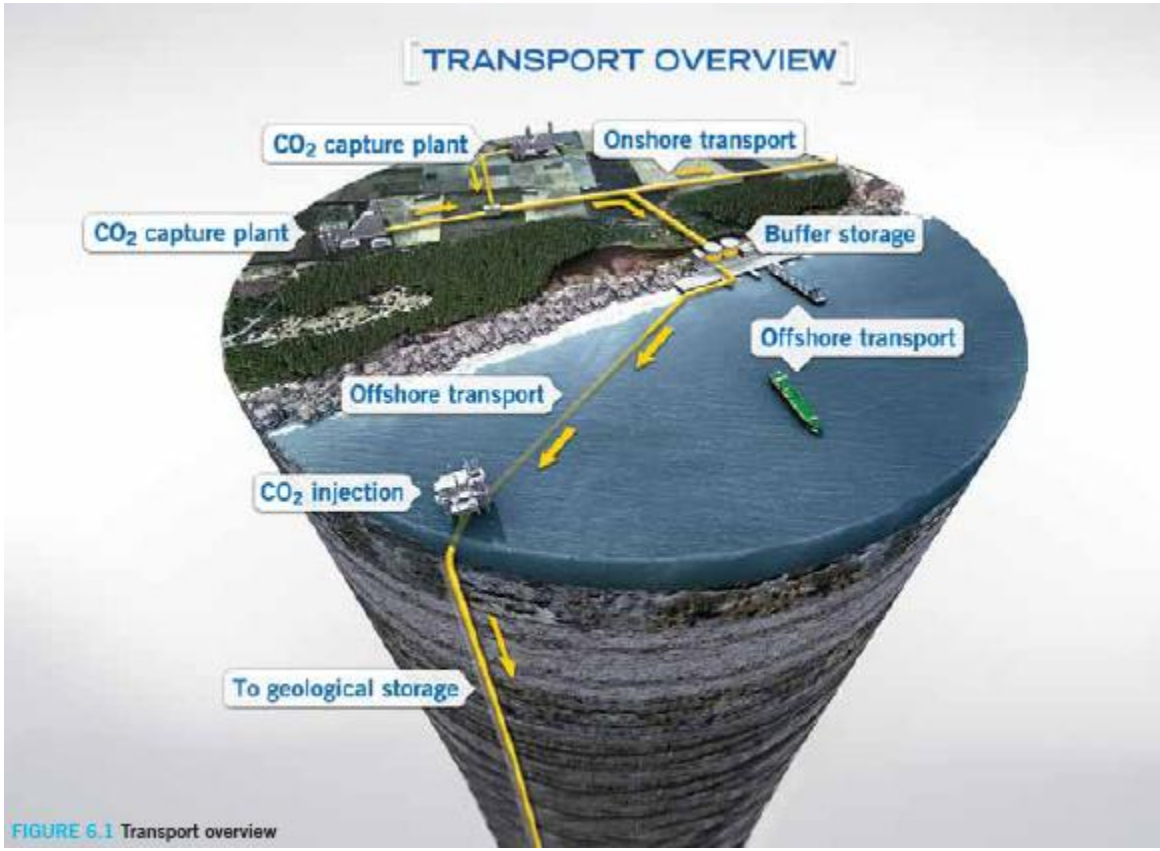


FIGURE 6.1 Transport overview

CO_2 transportation pathways and challenges

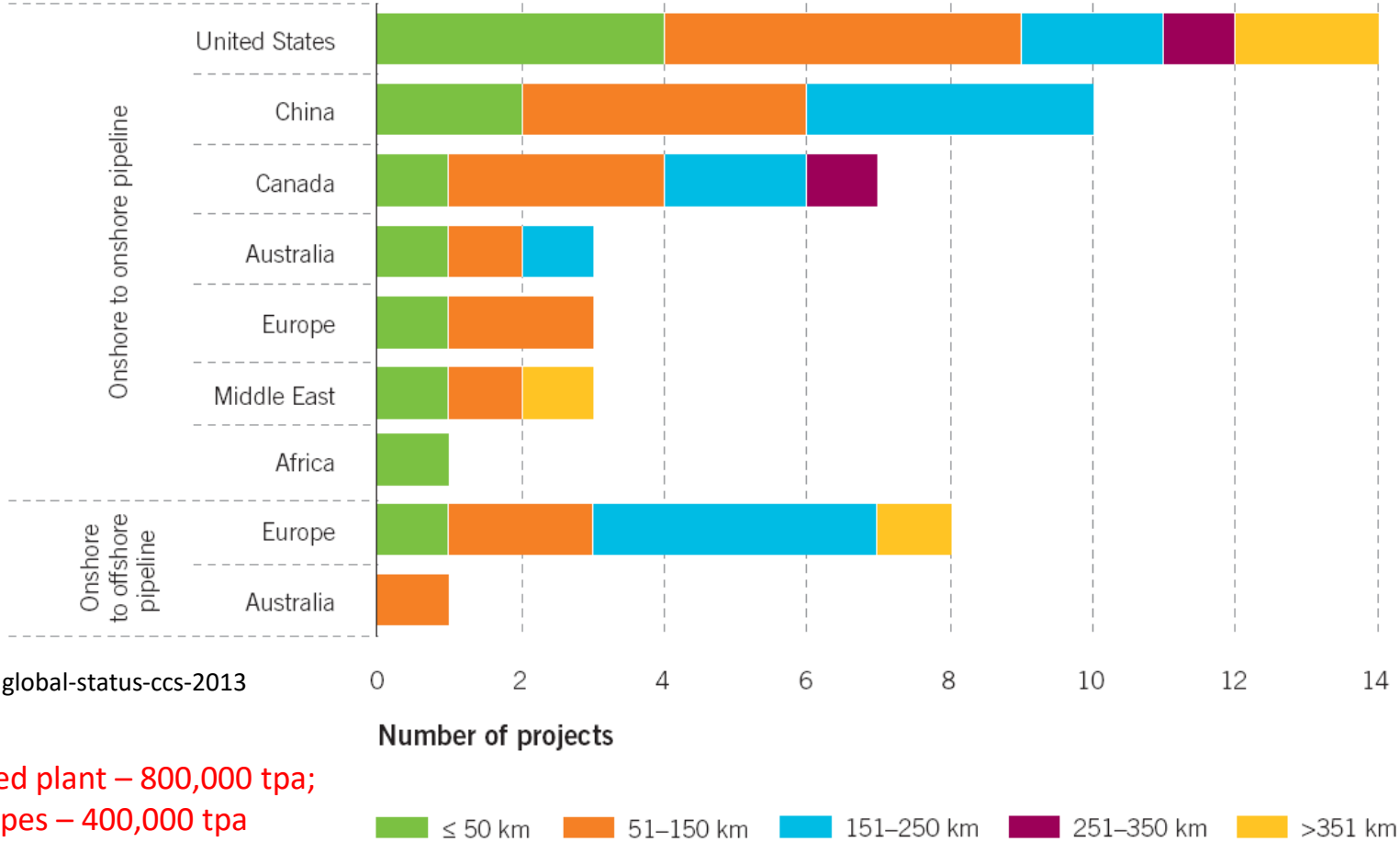
Research and technology implementation pathways

- CO_2 pipeline study
- Pipeline scenario modeling
- Facilitate communication between industry, academia, and government
- Impacts on materials from mixed gas streams

Challenges

- Public perception
- Regulatory uncertainty
- Uncertain CO_2 demand and price-economic feasibility
- Liability risks
- Large-scale integration of pipeline network
- Corrosion resistant alloys and coatings

Pipeline transportation distances provided by LSIPs



Source: global-status-ccs-2013

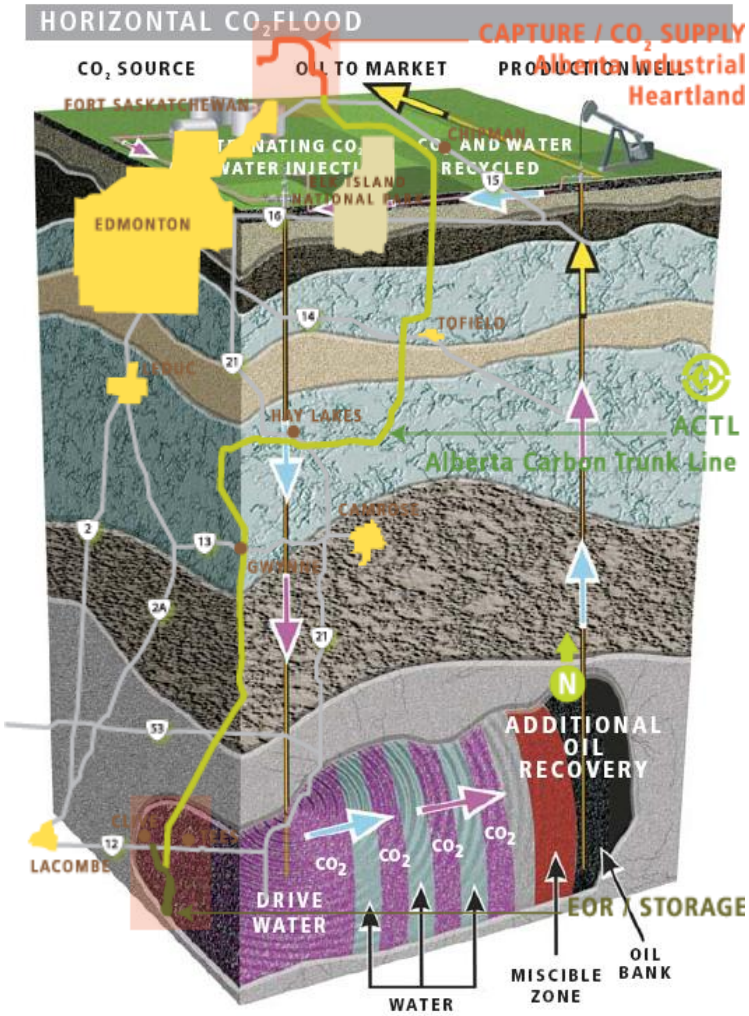
LISP:
Coal-fired plant – 800,000 tpa;
Other types – 400,000 tpa

The total length of pipeline transport to be developed for the 65 LSIPs will be around 6,000 km. This is approximately the same size as the existing network of dedicated CO_2–EOR pipelines in the US, which transports 48–58 Mtpa of CO_2 (DiPietro *et al., 2012). However, for CCS to contribute to meeting the* IEA 2DS, the estimated distance of CO_2 transportation infrastructure to be built in the coming 30–40 years is 100 times larger than currently exists.

EXPANDING CO_2 TRANSPORTATION NETWORKS

- Key milestones achieved in 2013: **Alberta Carbon Trunk Line** plans to develop shared CO_2 transport infrastructure
- With two projects moving into Execute and feeding into the pipeline, **the 240 km pipeline network** is becoming a reality
- At full capacity, the ACTL will compress, transport, and store up to 14.6 million tonnes of CO_2 /year

  = taking 2.6 million cars off the road annually



ACTL fact sheet from www.enhanceenergy.com

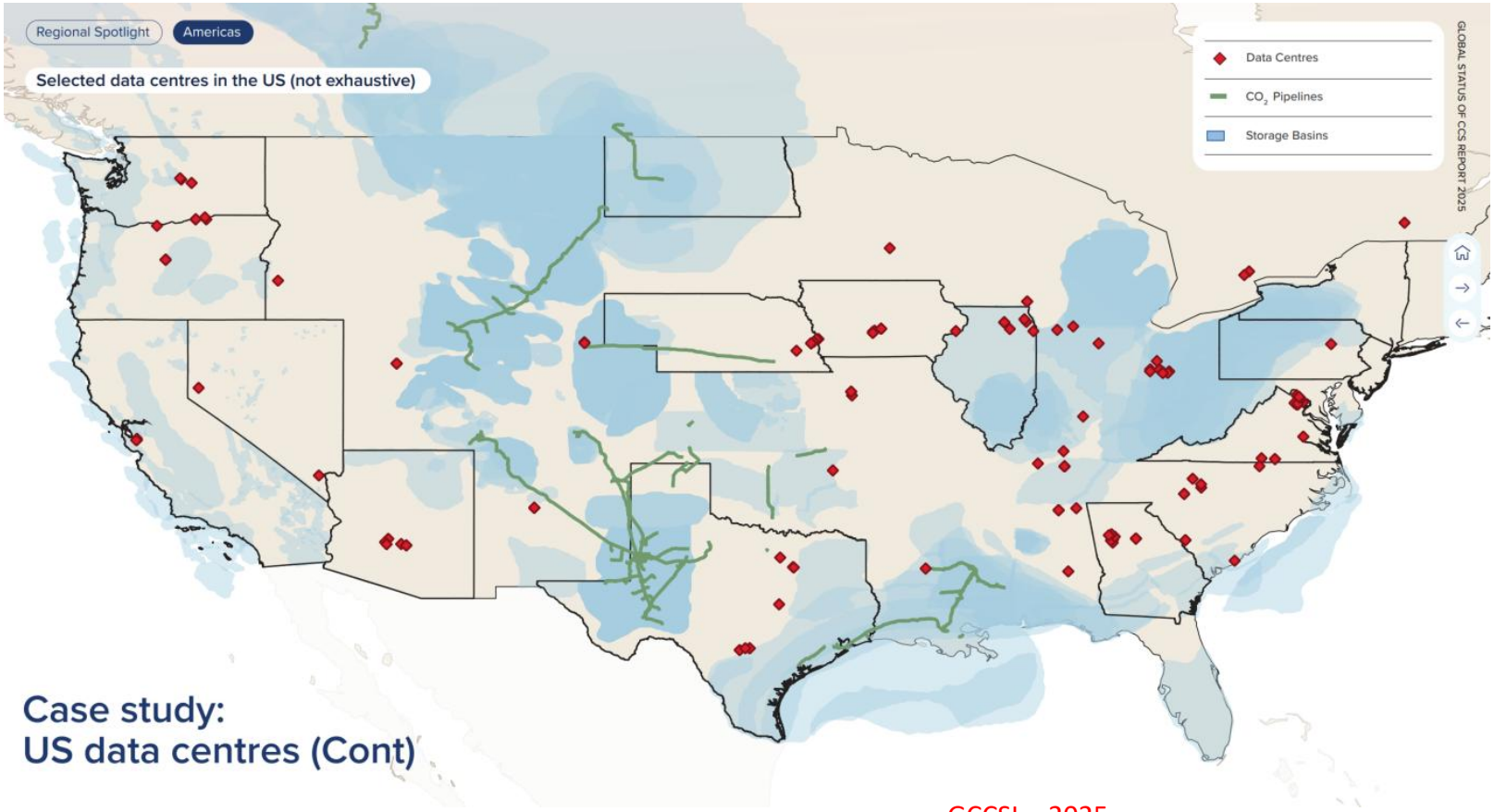
EXPANDING CO_2 TRANSPORTATION NETWORKS

- Policy： Prior to commencing the Alberta Carbon Trunk Line project, various local, provincial, and federal permits were acquired.
- Technology： The individual joints of pipe will be welded together to form 400–500 m sections, coated to minimise corrosion, and lowered into the ground.

EXPANDING CO_2 TRANSPORTATION NETWORKS

- In the US, much of the existing CO_2 pipeline infrastructure was built in the 1980s and 1990s and delivers mainly naturally sourced CO_2 for EOR purposes, rather than captured anthropogenic CO_2. However, there has been significant new investment over recent years. This includes the **373 km Greencore pipeline**, sanctioned at the end of 2012, and the **112 km Coffeyville – Burbank pipe line**, sanctioned in mid-2013.
- In the US, there are about 800,000 km of natural gas and hazardous liquid pipelines, and 3.5 million km of natural gas distribution lines. Some 6,000 km of pipelines actively transport CO_2 today.
- In the US, around 50 CO_2 pipelines are currently operating; in 2010, operational pipelines transported 48–58 Mtpa of CO_2 (DiPietro *et al., 2012). These onshore pipelines cross six provincial/*state boundaries and one international border (into Canada).

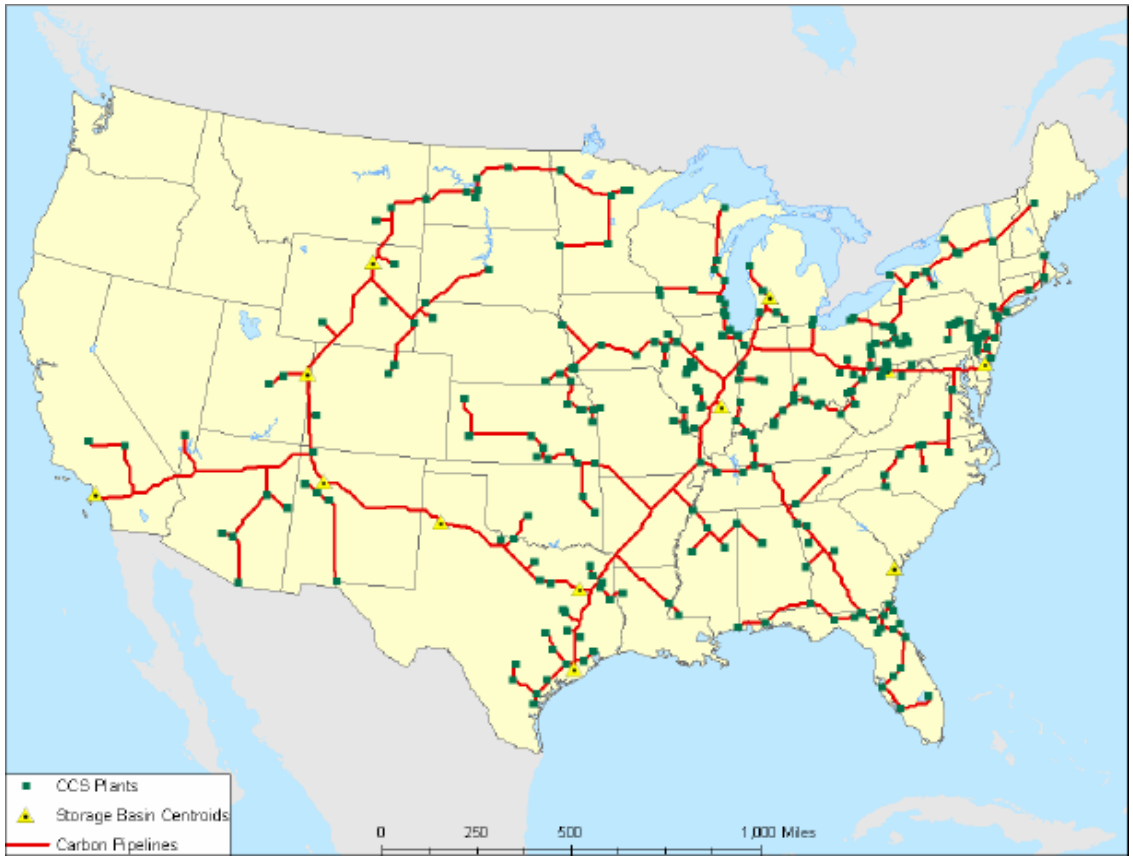
Location of US CO_2 pipelines



GCCSI，2025

Some 6,000 km of pipelines actively transport CO_2 today. and currently operational pipelines transported 48–58 Mtpa of CO_2 each year.

A Possible Carbon Dioxide Trunkline System for the US



- NB: It may be noted that emissions of carbon dioxide from power plants represents some 40% of the total generated within the boundaries of the United States. Additional sources include the consumption of petroleum, natural gas and coal by the transport (34% !!) commercial and industrial (20%) and residential (6%) sectors.

EXPANDING CO_2 TRANSPORTATION NETWORKS

- Australia: two projects have made progress on their 'hub concept'.

CO_2 NETWORK PROPOSALS FOR CCS	DESCRIPTION AND ANCHOR LSIPS
South West Hub	The South West CO_2 Geosequestration Hub project in Western Australia aims to be collecting 5–6 Mtpa of CO_2 by 2018–22 from industrial processes, including the Perdaman Collie urea project, as well as from alumina production and power facilities for storage in the Lesueur formation in the Southern Perth Basin.
CarbonNet Project	The CarbonNet CCS network aims to integrate multiple CCS projects across the entire CCS value chain within the next 10 years. The network is initially sized to capture and store around 1 Mtpa of CO_2 from emission sources in the Latrobe Valley by 2018, with the potential to rapidly scale up to support more than 20 Mtpa thereafter.

EXPANDING CO_2 TRANSPORTATION NETWORKS

- The Middle East: the Masdar project has completed its pipeline FEED study and is working toward an FID in order to connect various sources of CO_2 in the UAE to an onshore EOR site.

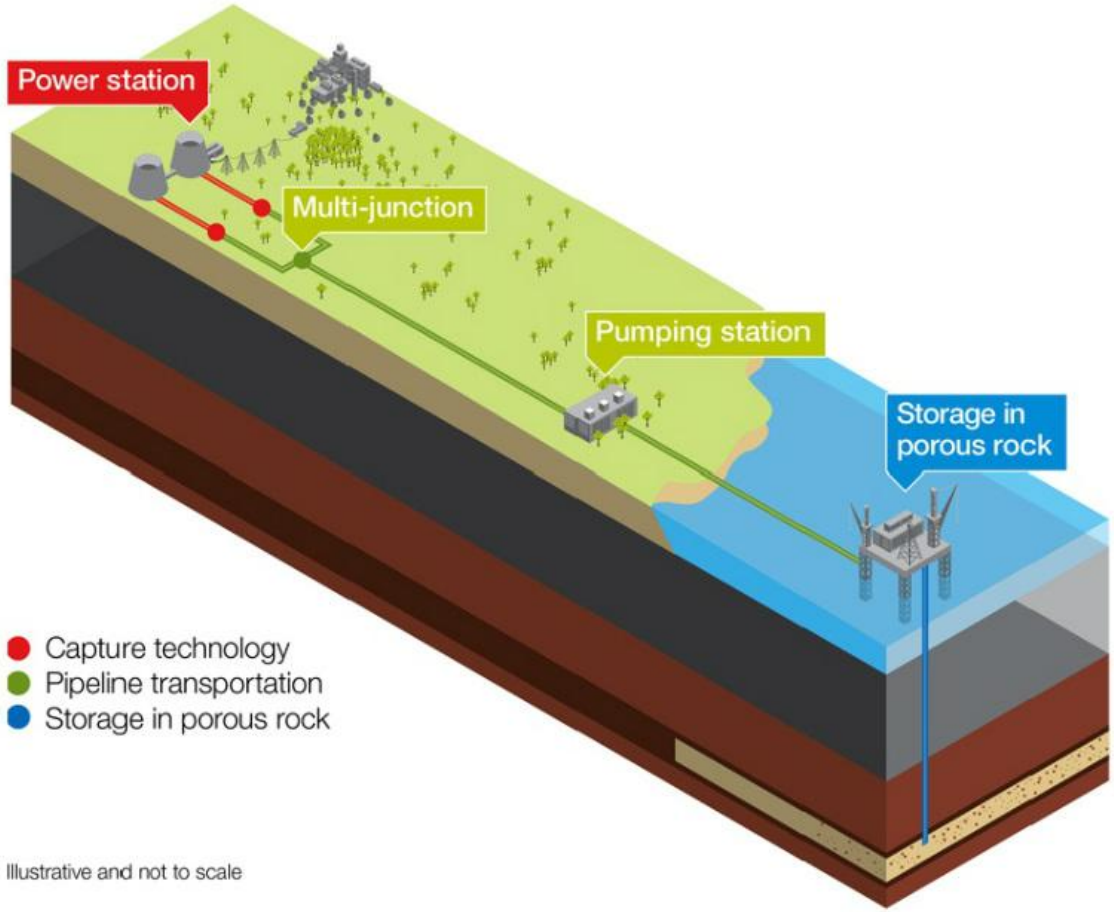
CO2 NETWORK PROPOSALS FOR CCS	DESCRIPTION AND ANCHOR LSIPS
Masdar CCS Project (UAE)	The Abu Dhabi CCS network (Masdar) aims to capture CO_2 emissions from existing power and industrial sites, as well as develop a network of CO_2 pipelines to transport the CO_2 to Abu Dhabi's oil reservoirs for EOR. Anchor projects include ESI CCS Project and the Emirates Aluminium CCS Project. By 2014, Masdar would have captured a total of five million tonnes of CO_2.

EXPANDING CO_2 TRANSPORTATION NETWORKS

- Europe: the focus has changed from realising ambitious CO_2 networks (in which CO_2 from multiple sources is collected and transported by a shared infrastructure system to one or more CO_2 storage or EOR sites) to getting individual projects 'over the line'.

CO_2 NETWORK PROPOSALS FOR CCS	DESCRIPTION AND ANCHOR LSIPS
Rotterdam CO2 Hub (The Netherlands)	The Rotterdam CO2 Hub aims to capture and store 5 Mtpa of CO2 from anchor projects like ROAD and other industries in the port in 2015–17, expanding to 20 Mtpa in 2020–25 and providing the basis for low–carbon industrial and economic growth in Rotterdam.
Humber cluster (UK)	The Humber and Yorkshire region has the long-term potential to capture and store upwards of 40 Mtpa of CO2 from many sources. Anchor projects include the White Rose Oxyfuel project, C.GEN North Killingholme Power Project, and Don Valley Power Project.
Teesside cluster (UK)	The cluster in the Teesside region would capture and store up to 15 Mtpa of CO2 from the Teesside Low Carbon project, an aluminium smelter, and emissions from other surrounding industries.
Scottish CCS cluster (UK)	The Captain Clean Energy Project could accelerate the development of a Scottish CCS cluster. The CO_2 captured in the Firth of Forth area will be transported by pipeline to the St Fergus terminal in close proximity to SSE's Peterhead project, where CO_2 Deep Store will store it in depleted reservoirs under the North Sea.

Yorkshire and Humber CCS Project



https://www.gov.uk/government/news/drax

Offshore CO_2 transportation – shipping

Transportation by ship may be an alternative to pipelines in some parts of the world, particularly in regions where there is limited access to nearby CO_2 storage reservoirs.

Three active LSIPs are currently considering shipping their CO_2 to offshore storage locations. These include the two CCS projects developed by the Korean Electric Power Corporation, and the Dongguan Taiyangzhou IGCC with CCS Project in the Chinese province of Guangdong.

Taiyangzhou IGCC 750 MW (netplant

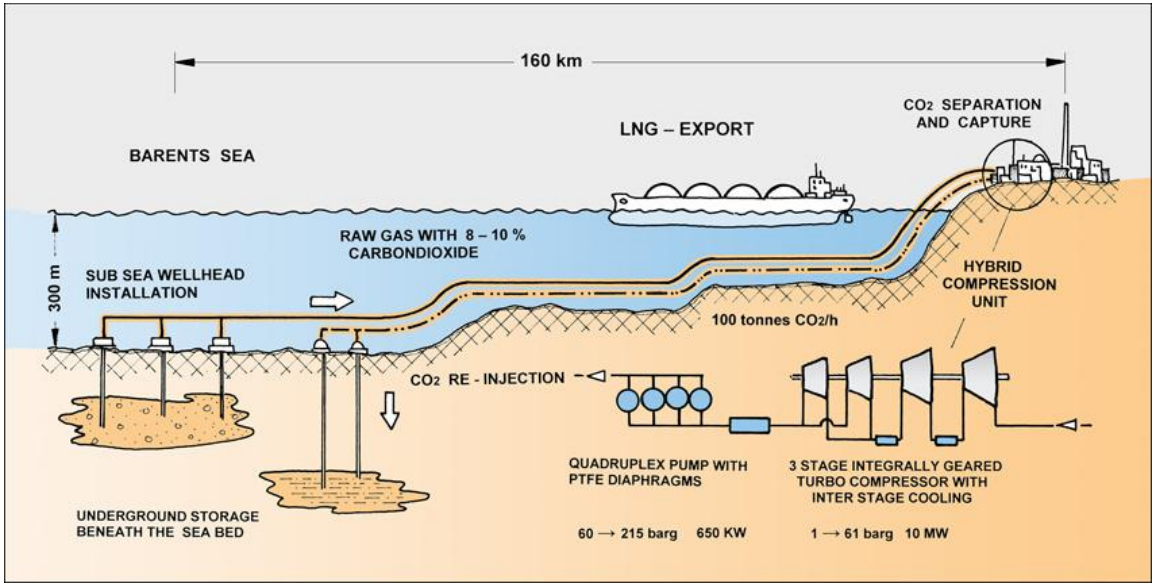
Amount:1 million t/y
The CO_2 would be transported in a pipeline and stored in near-depleted offshore oil and gas reservoirs about 100 km from the power plant.

Offshore CO_2 transportation-EU

Several European projects around the North Sea are considering CO_2 ships as a secondary transport option, but the main focus is on offshore pipelines. The only offshore pipeline for CO_2 currently in use is part of the Snøhvit project (Norway), which has been operational since 2008 and covers some 153 km to link Hammerfest to the Snøhvit field under the Barents Sea.

In April 2008, Statoil announced carbon storage had started , 5% to 8% of CO_2 is removed from the gas and piped back to a 2,600m-deep sandstone formation at Snøhvit, where it sits under the seabed. The process will reduce CO_2 emissions by 700,000t a year when Snøhvit is at full capacity, it is estimated. This is the equivalent of emissions from 280,000 cars.

New offshore pipelines are being considered by projects in The Netherlands, Norway, and UK to transport CO_2 via pipeline to various offshore storage locations in the North Sea.



Pipeline design – the ROAD project

This section will share insights into pipeline design and CO_2 stream operation philosophy from one of the first LSIPs in the power sector, ROAD in The Netherlands. The ROAD pipeline system is 25 km long, of which 20 km is offshore. Approximately 1.1 Mtpa.

Pipeline design – the ROAD project

Design parameters and data

TABLE 6.1 Impurities in the CO_2 stream in parts per million by volume (ppmv)

NITROGEN (N_2)	WATER (H_2O)	OXYGEN (O_2)	ACETALDEHYDE (C_2H_4O)	ARGON (Ar)
≤350 ppmv	≤50 ppmv	≤50 ppmv	≤10 ppmv	≤7 ppmv

Impurities or by-products may change the density of the CO_2 stream, which could, for example, affect the pressure needed to transport CO_2 in a dense phase (as opposed to a gaseous phase). Moreover, combinations of impurities might influence the physical properties of the CO_2 stream differently to impurities from a single component.

At every stage of the process, the only variables available to the operator are the temperature of the CO2 entering the pipeline (in the range of 40–80° C, limited by cooling water availability) and the absolute pressure (in the range 40–125 bar, determined by the compressor).

Pipeline design – the ROAD project

- The FAS carried out for the ROAD project has shown that under most of the operating conditions the flow in the pipeline is homogeneous and stable.
- In the first phase of the operation (for well pressures from 30–73 bar) the CO_2 is transported in the gas phase and the required pipeline entrance temperatures run initially at 80° C, decreasing to 60° C. As the reservoir pressure increases, to avoid slugs, the CO2 temperature at the pipeline entrance should be below 60° C and up to 200 bar reservoir pressure, decreasing to 40° C at an absolute reservoir pressure of 300 bar.
- To lower the required operating pressure and save considerable amounts of compression energy, a control valve at the injection platform has been used to increase pipeline pressure, which will force the CO_2 flow into single phase operation, thus avoiding two-phase flow and unstable slugs.
- SYSTEM FLEXIBILITY – START-UP AND SHUTDOWN: The CO_2 capture and storage system of ROAD are largely governed by economic parameters, so it is likely that frequent start-ups and shutdowns will be necessary.

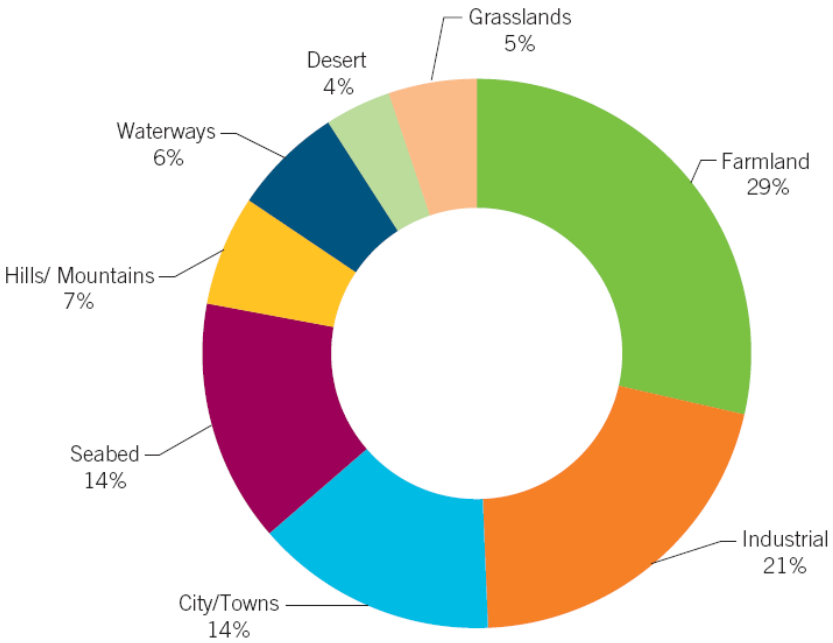
Permitting CO_2 transport infrastructure – standards and codes

The 2013 survey of LSIPs revealed that, on average, it takes a project between one and two years to work through the permitting process. The time it takes to get a CO_2 pipeline permitted correlates strongly to the terrain a transport pipeline needs to cross. 14 per cent of LSIPs operate, or plan to operate, a pipeline that traverses cities and towns, whereas the majority of projects only affect farmland and industrial areas.

CO_2 transportation experience:

- ASME B31.4 Pipeline Transportation Systems for Liquid Hydrocarbons and Other Liquids(2006)
- Canadian Standards Association's Z662-07 standard for Oil and Gas Pipeline Systems(2011).
- Australian Standard – AS 2885: Pipelines – Gas and Liquid Petroleum (2012).
- The DNV Joint Industry Project publication 'Recommended Practice: Design and operation of CO2 pipelines' addresses the gaps in existing standards (DNV, 2010), in particular the ISO standard 13623: Petroleum and Natural Gas industries – Pipeline Transportation Systems (2009).

Pipeline terrain for LSIPs



International standards for CO_2 pipelines

- In May 2011, the Standards Council of Canada (SCC) submitted a proposal to the ISO to develop an internationally agreed and voluntary standard for CCS. The ISO subsequently agreed to pursue a proposed program **(ISO/TC 265)** of work that includes the full lifecycle of a CCS system, including CO_2 transportation.
- The working group on CO_2 transport (IS/TC 265/ WG2 'CO_2 Transportation') met for the first time in June 2013 and defined the scope of work as follows.
- The work on international standards for CO_2 pipelines is completed in 2016. (**ISO 27913:2016** Carbon dioxide capture, transportation and geological storage — Pipeline transportation systems)

- *Based on existing standards for transportation of gaseous or liquid media the new CO2 standard will define additional requirements or recommendations for CO2 transportation by pipelines. Special care has to be taken in the design phase to select the right material, wall thickness, internal corrosion protection etc. Construction, testing, operation, maintenance and abandonment of pipeline systems used for transportation of CO2 have also to be taken into account. This International Standard will include aspects of CO2 stream quality assurance as well as converging CO2 streams from different sources. There is also a need to specify the purity, composition and concentration of the CO2 stream delivered to the pipeline system boundary. It is also important to take into account the thermodynamics and phase behavior of the CO2 stream during design and operation of the pipeline system. Health, safety and environment specific to CO2 transport and monitoring will also be considered.*

ISO 27913:2016

Carbon dioxide capture, transportation and geological storage — Pipeline transportation systems

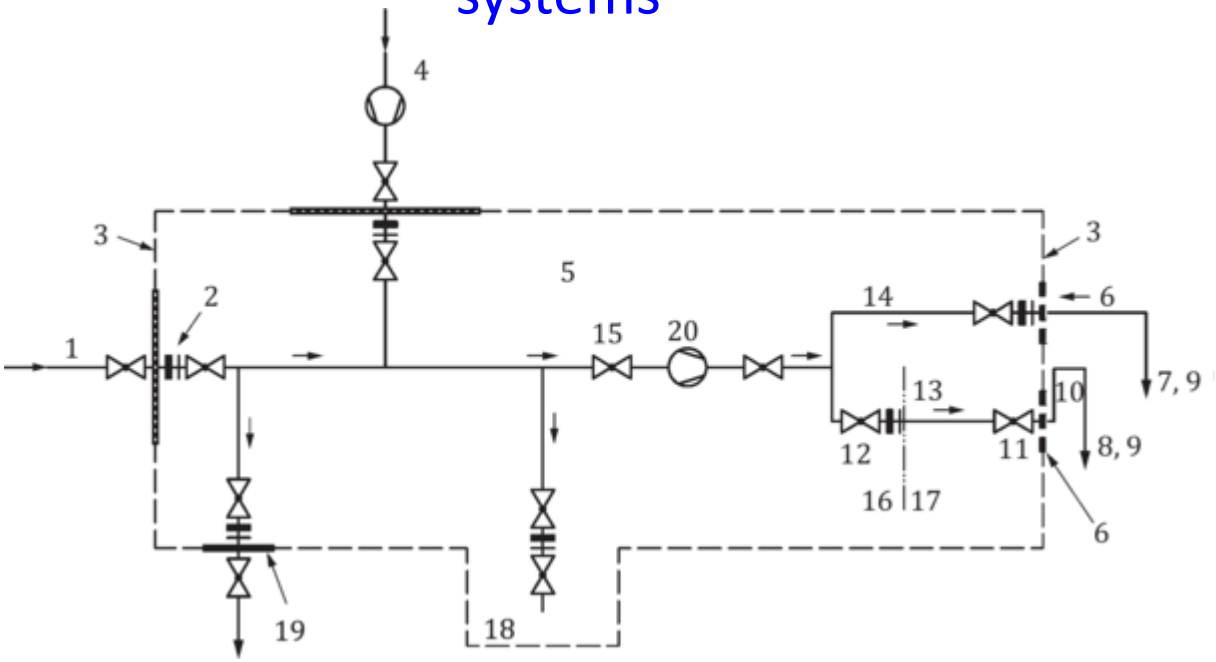
ISO 27913:2016 specifies additional requirements and recommendations not covered in existing pipeline standards for the transportation of CO_2 streams from the capture site to the storage facility where it is primarily stored in a geological formation or used for other purposes (e.g. for EOR or CO_2 use).

ISO 27913:2016 applies to

- **rigid metallic pipelines,**
- **pipeline systems,**
- **onshore and offshore pipelines for the transportation of CO_2 streams,**
- **conversion of existing pipelines for the transportation of CO_2 streams,**
- **pipeline transportation of CO_2 streams for storage or utilization,**
- **transportation of CO_2 in the gaseous and dense phases.**

ISO 27913:2016

Carbon dioxide capture, transportation and geological storage — Pipeline transportation systems



1 source of CO_2 from capture, e.g. from power plant,
industry; see ISO/TR 27912 (capture)
2 isolating joint
3 boundary limit
4 other source of CO_2
5 ISO 27913 (transportation system inside)
6 boundary to storage facility
7 onshore storage facility
8 offshore storage facility
9 EOR
10 riser (out of transport scope)
11 subsea valve (inside transport scope)
12 beach valve
13 offshore pipeline
14 onshore pipeline
15 valve
16 landfall
17 open water/sea
18 third party transport system
19 export to other uses than 7, 8 and 9
20 intermediate compression or pumping

The system boundary between capture and transportation is the point at the inlet valve of the pipeline, where the composition, temperature and pressure of the CO_2 stream is within a certain specified range by the capture process or processes to meet the requirements for transportation as described in this document.
The boundary between transportation and storage is the point where the CO_2 stream leaves the transportation pipeline infrastructure and enters the storage infrastructure.
ISO 27913:2016 also includes aspects of CO_2 stream quality assurance, as well as converging CO_2 streams from different sources.
Health, safety and environment aspects specific to CO_2 transport and monitoring are considered.

TECHNICAL COMMITTEES

ISO/TC 265

Carbon dioxide capture, transportation, and geological storage

ABOUT

SECRETARIAT: SCC 加拿大 Canadian Standards Association

Committee Manager: Ms Lynn Barber

TWINNED SECRETARIAT: SAC 中国

Twinned Committee Manager: Ms Mei LIU

Chairperson (until end 2023): Mr William (Bill) Spence

ISO Technical Programme Manager [TPM]: Mrs Kirsi Silander - van Hunen
ISO Editorial Programme Manager [EPM]: Ms Jasmin Lewis

Creation date: 2011

ISO/TC 265 - TWINNED SECRETARIAT

SAC [CHINA]

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SCOPE

Standardization of design, construction, operation, environmental planning and management, risk management, quantification, monitoring and verification, and related activities in the field of carbon dioxide capture, transportation, and geological storage (CCS).

国际标准化组织（ISO）二氧化碳捕集、运输与地质封存（CCS）标准化技术委员会（TC265）于2012年成立，范围覆盖了二氧化碳捕集、运输，二氧化碳地质封存，量化验证，交叉问题等领域。中国标准化研究院和加拿大标协联合承担该委员会秘书处，资源与环境分院刘玫研究员担任联合秘书。

PARTICIPATING MEMBERS (19)

COUNTRY/TERRITORY	ACRONYM
Australia	SA
Canada	SCC
China	SAC
France	AFNOR
Germany	DIN
India	BIS
Italy	UNI
Japan	JISC
Korea, Republic of	KATS
Luxembourg	ILNAS
Malaysia	DSM
Netherlands	NEN
Norway	SN
Portugal	IPQ
Saudi Arabia	SASO
South Africa	SABS
Spain	UNE
United Kingdom	BSI
United States	ANSI

OBSERVING MEMBERS (13)

COUNTRY/TERRITORY	ACRONYM
Argentina	IRAM
Czech Republic	UNMZ
Denmark	DS
Egypt	EOS
Finland	SFS
Hungary	MSZT
Iran, Islamic Republic of	ISIRI
Mexico	DGN
New Zealand	NZSO
Qatar	QS
Serbia	ISS
Sri Lanka	SLSI
Sweden	SIS

9 PUBLISHED ISO STANDARDS *
under the direct responsibility of ISO/TC 265

STANDARD AND/OR PROJECT UNDER THE DIRECT RESPONSIBILITY OF ISO/TC 265 SECRETARIAT (9)

ISO/TR 27912:2016
Carbon dioxide capture — Carbon dioxide capture systems, technologies and processes

ISO 27913:2016
Carbon dioxide capture, transportation and geological storage — Pipeline transportation systems

ISO 27914:2017
Carbon dioxide capture, transportation and geological storage — Geological storage

ISO/TR 27915:2017
Carbon dioxide capture, transportation and geological storage — Quantification and verification

ISO 27916:2019
Carbon dioxide capture, transportation and geological storage — Carbon dioxide storage using enhanced oil recovery (CO2-EOR)

ISO 27917:2017
Carbon dioxide capture, transportation and geological storage — Vocabulary — Cross cutting terms

ISO/TR 27918:2018
Lifecycle risk management for integrated CCS projects

ISO 27919-1:2018
Carbon dioxide capture — Part 1: Performance evaluation methods for post-combustion CO2 capture integrated with a power plant

ISO/TR 27921:2020
Carbon dioxide capture, transportation, and geological storage — Cross Cutting Issues — CO2 stream composition

ISO/TR 27921:2020(en) Carbon dioxide capture, transportation, and geological storage — Cross Cutting Issues — CO2 stream composition

- Depending among other things on the feedstock and the CO_2 generating and capture processes, CO_2 streams captured from industrial sources or power generation contain various impurities (that is, stream components other than CO_2). The impurities differ in their concentrations but also in their physical and chemical properties. Therefore, the composition of the originally captured **CO_2 stream is a key starting point in ensuring the safety and reliability of the transport and geological storage of CO_2**. Compositional information may assist operators in evaluating the need for treating a CO_2 stream, based on the intended transportation options (including mixing with other CO_2 streams), usage options (EOR or other), or dedicated storage in geologic formations.
- Understanding CO_2 stream composition is also **important for the commerciality of CCS activities** because additional purification of the CO_2 stream increases CO_2 capture costs. In addition, CO_2 stream composition is an **important input factor in quantifying the amount of CO_2 stored**, for the purpose of greenhouse gas emissions reporting and crediting. Because capture and purification processes are continuously evolving, it is relevant to expose the range of impurities in CO_2 streams and their concentrations, including recent publications.
- This document provides up-to-date consideration of CO_2 stream quality issues for operators, regulators and stakeholders based on research results and the experience of various pilot and industrial scale CCS projects. The first part of this report summarises **existing information related to CO_2 stream composition that generally results from particular capture processes**. Then this report **describes possible impacts and effects of the various impurities** that may occur in the CO_2 stream on various downstream elements of a CCS chain, including operational aspects, potential implications for health, safety and environmental issues, and quantification of greenhouse gas emissions.

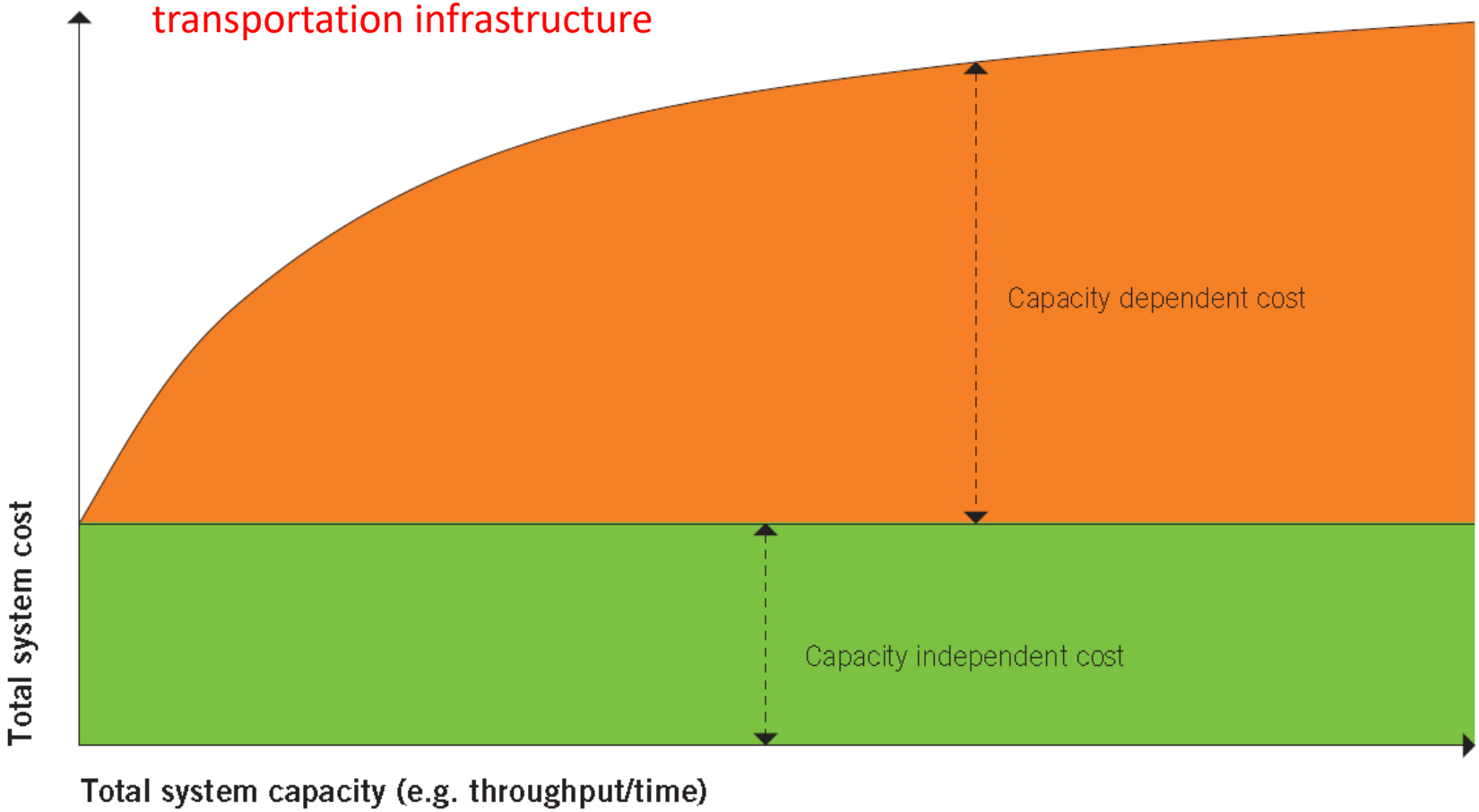
Financial and commercial models for CO_2 transportation networks

- CO_2 transport costs may be in the order of two to five per cent of the investment needed for a complete CCS facility, but they are still significant in the demonstration phase, ranging from US$2–7 per tonne of CO_2 for transport up to 200 km.

- expected volumes of CO_2
- the corresponding pipe diameters
- cost of labour
- economic life of the infrastructure

Financial and commercial models for CO_2 transportation networks

Cost allocation approaches to multi-user CO_2 transportation infrastructure



Share transport and storage infrastructure could lead to significant cost savings

Recommendation and outlook

- Global sharing of pipeline design, construction, and operation experience through industry best practice guidelines and standards

  全球共享管道设计、施工、实践准则和标准经验

- Further R&D and demonstration of CO_2 shipping concepts, as well as engineering challenges imposed by new operating requirements and more challenging pipeline terrains

  二氧化碳运输的概念研发和示范，新地形的要求和挑战性

- Development and implementation of appropriate international standards and design codes to further promote safe and efficient operation of CO_2 transport infrastructure

  国际标准和设计规范推动二氧化碳运输基础设施的安全高效运行

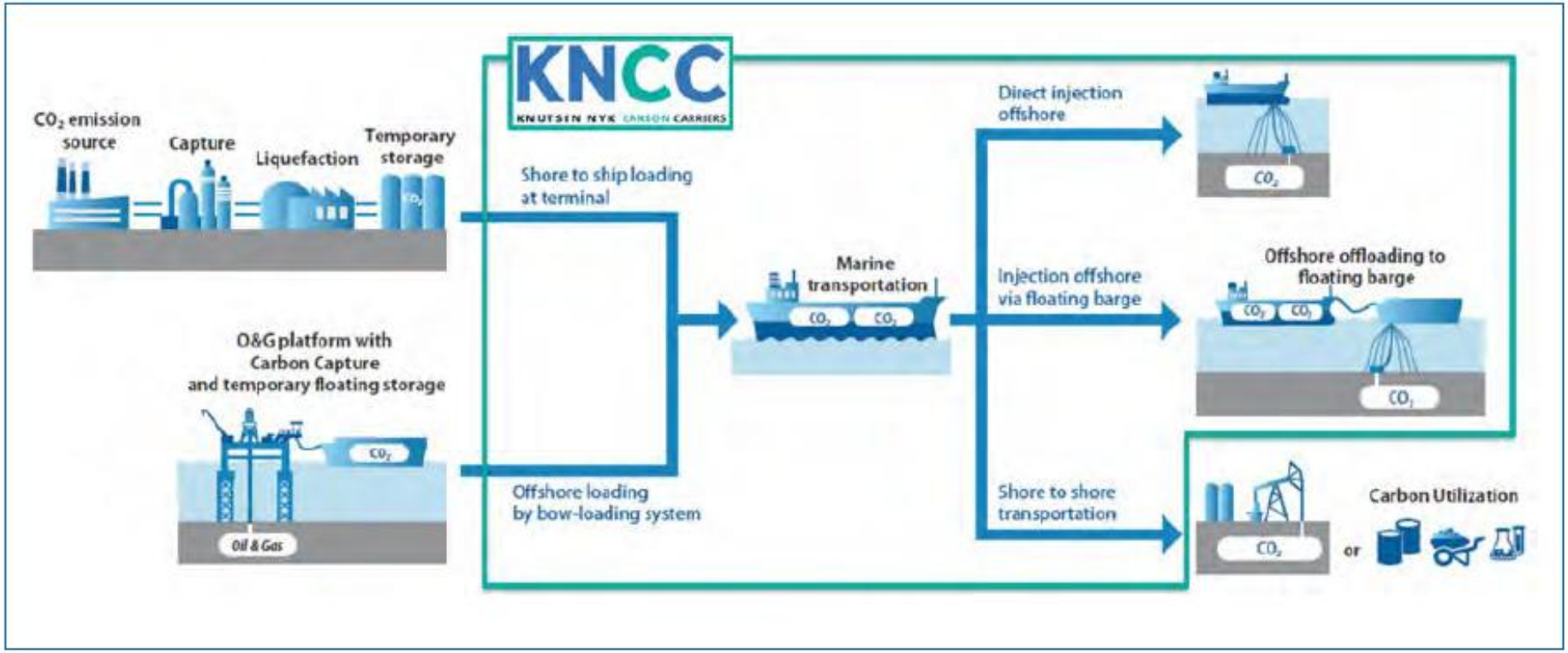
- Implementation of adequate incentives for projects to invest in oversized CCS transport solutions that will accommodate the future development of other CCS projects and large CO_2 transportation networks.

  适当激励大投资CCS运输方案，容纳其他CCS和大型CO_2的运输网络

Ship transport

Kawasaki Kisen Kaisha, Ltd.川崎汽船株式会社(“K” LINE), founded in 1919, is one of the largest shipping companies in the world. Firstly, since 2021, “K” LINE is participating in the demonstration project on LCO_2 transportation supported by New Energy and Industrial Technology Development Organization (NEDO), Japanese national research and development agency.

Knutsen NYK Carbon Carriers AS（KNCC）



- 日本邮船与挪威航运公司Knutsen Group成立合资企业，将在全球范围内进行液化二氧化碳（LCO2）海上运输和储存业务的商业开发。**LCO2船则将在CCUS价值链中发挥重要作用**.
- PCO2储罐系统能够在环境温度（0-10摄氏度）和高压（35-45巴）的条件下运输LCO2，相比低压或中压运输概念可以实现更大船型规模的LCO2运输，降低整个碳捕获、利用和封存（CCUS）价值链的成本，获得船级社颁发认证。

碳运输：

我国 CCUS 项目正在由小规模、先导式向规模化、集群化示范转型，配套的二氧化碳运输技术也在向更大规模、更长距离、更经济的方向发展。相关运输装备、基础设施在近几年得到推动。

01.海上二氧化碳运输

全球首艘商用液态二氧化碳运输船交付

- 2024 年 11 月，大连船舶重工集团有限公司成功交付了全球首艘商用 7500 立方米液态二氧化碳运输船“北极光先锋”号。该船配置有两个特殊材料制造的全压式 C 型液货罐，可耐 -35℃低温，用于运输二氧化碳[34]。

液态二氧化碳运输船

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[32] 齐耀环保. 首台套船用碳捕集系统关键设备顺利通过 FAT 验收[EB/OL]. (2023-07-21) [2025-03-11]. http://www.qiyaoetech.com/news_detail/32.html

[33] 中远海运重工有限公司. 全球首艘具有碳捕集功能 FPSO 在上海中远海运重工命名[EB/OL]. (2025-02-20) [2025-03-11]. https://chi.coscoshipping.com/col/col7982/art/2025/art_c39aa55f56a94231908a5a507caf067f.html

[34] 中国船舶集团有限公司. 冲刺四季度丨全球首制！商用 7500 立方米液态二氧化碳运输船交付[EB/OL]. (2024-12-02) [2025-03-11]. http://www.csic.com.cn/n135/n173/n193/c30801/content.html

02.运输管道及材料

各类尺寸的二氧化碳运输管道产品

- 管线管作为输送二氧化碳的管道材料，是 CCUS 项目输送二氧化碳的重要组成部分。2024 年 12 月，包钢股份钢管公司研发生产的二氧化碳输送管线管产品实现首次供货，为包钢 200 万吨（一期 50 万吨）CCUS 全产业链示范项目定制，具有优良的冲击韧性。

二氧化碳管道产品生产

- 2024 年 12 月，中国石油渤海装备研制了直径 610 毫米、壁厚 38 毫米的超临界二氧化碳输送用直缝埋弧焊管产品，是我国目前最大管径、最大壁厚的超临界二氧化碳输送用直缝埋弧焊管[35]。与此同时，中油宝世顺公司与钢管研究院也联合开发了最小口径且低温性能优异的 L450M 钢级超临界二氧化碳输送用直径 355 毫米、壁厚 12.7 毫米的 JCOE 直缝埋弧焊管，可满足我国小输量输送超临界二氧化碳用管的迫切需求[36][37]。

百公里二氧化碳运输管道投运

- 2023 年 7 月，“齐鲁石化-胜利油田百万吨级 CCUS 项目”二氧化碳输送管道正式投运。管道全长 109 公里，是我国首条百万吨输送规模、百公里输送距离、百公斤输送压力的高压常温密相二氧化碳输送管道。该项目是我国首次实现液体二氧化碳长距离密相管输，对推动 CCUS 全产业链规模化发展具有重要意义[38]。

齐鲁石化 CO_2 管道

- 例如，管道投用前，项目二氧化碳运输依赖于罐车，每辆车装载约 25.5 吨二氧化碳，往返一趟需要 7 个多小时，运输成本相对较高。二氧化碳管道运输投用后，做到捕集、输送、利用、封存全过程、全密闭，每年可减少 4 万辆罐车行驶百公里运输费用，降低了交通资源占用和二氧化碳排放[39]。

400 公里二氧化碳运输管道规划建设

- 吉林油田正推进国内更大规模的二氧化碳输送管道建设。设计的吉林石化—吉林油田二氧化碳长输管道总长约 400 公里，途经长春、吉林、松原、四平 4 市 12 个县区，分两期建设[40]。
- 长输管道有望支持将吉林石化捕集的二氧化碳，输送至吉林油田用于驱油并埋存，实现碳减排与油增产的双赢。项目建成后有望成为吉林省各碳排放企业与利用企业的主干碳网，形成覆盖吉林省的碳产业链，极大促进吉林地区企业绿色低碳转型和高质量发展。